THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – III : Second Law of Thermodynamics

★ Note: Syllabus: Kelvin-Planck & Clausius statements, equivalence, PMM-II, Carnot's principle & cycle, thermodynamic temperature scale, Clausius inequality, entropy, principle of entropy increase, availability & irreversibility, Gibbs & Helmholtz functions, Maxwell relations, third law

1. Need for the Second Law

The first law only balances energy. It cannot say which way a process occurs, nor how much heat can be turned into work. Observations:

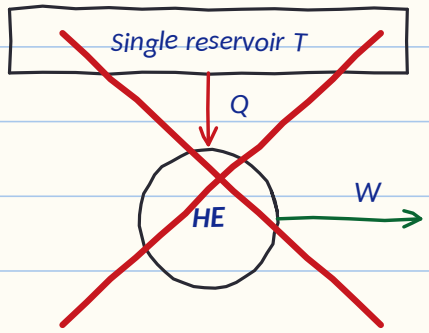
- Heat flows from hot to cold, never spontaneously from cold to hot.
- Work can be completely converted into heat (friction), but heat cannot be completely converted into work in a cycle.
- Gas expands into vacuum but never compresses itself back; mixing is never un-mixed by itself.

★ Note: The second law gives the direction of processes, the quality of energy, and the maximum possible efficiency of heat engines.

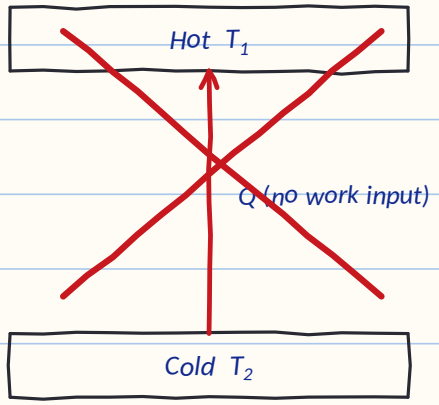
2. Statements of the Second Law

★ Note: Kelvin–Planck statement: It is impossible for a device operating in a cycle to receive heat from a single reservoir and produce a net amount of work.

★ *Note:* Clausius statement: It is impossible to construct a device operating in a cycle whose sole effect is the transfer of heat from a cooler body to a hotter body.



Kelvin–Planck violated

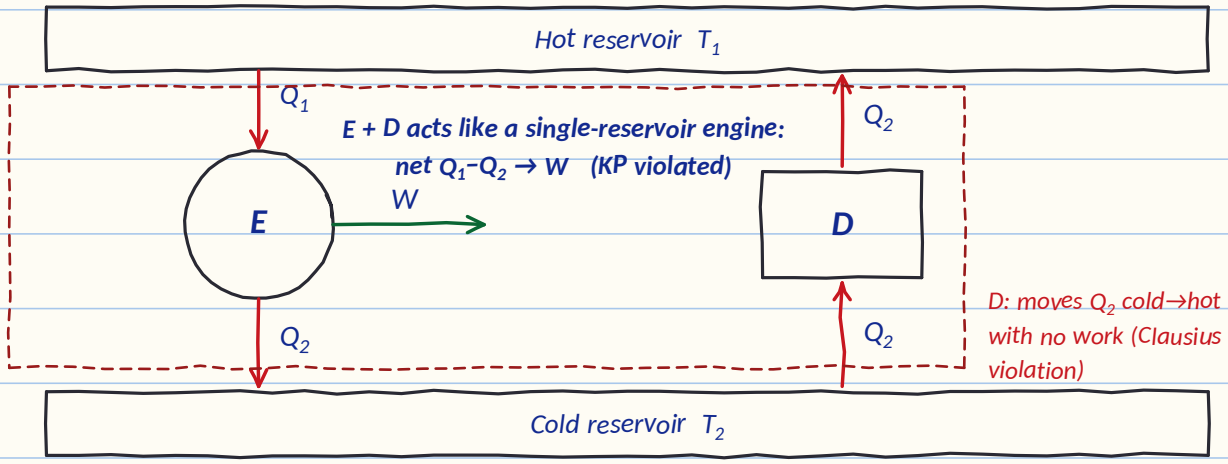


Clausius violated

- K-P: a heat engine must reject some heat to a sink ⇒ thermal efficiency is always less than 100 %.
- Clausius: a refrigerator needs work input; COP_R cannot be infinite.
- K-P deals with heat engines; Clausius with refrigerators / heat pumps. Both are negative statements – proved by the absence of any contrary experience.

3. Equivalence of Kelvin–Planck and Clausius Statements

The two statements are equivalent: violating one means violating the other.



(a) Violation of Clausius ⇒ violation of Kelvin–Planck

→ Let device D move Q_2 from cold (T_2) to hot (T_1) with no work (violates Clausius).

→ Engine E takes Q_1 from hot reservoir, produces W and rejects Q_2 to the cold reservoir.

→ Combined (E + D): the cold reservoir is unchanged; the hot reservoir loses $Q_1 - Q_2$ and a net work $W = Q_1 - Q_2$ is produced from a single reservoir ⇒ K-P violated.

(b) Violation of Kelvin–Planck ⇒ violation of Clausius

→ Let engine X take Q from hot reservoir and convert it fully to work $W = Q$ (violates K-P).

→ Use W to drive a refrigerator R: it takes Q_2 from cold and rejects $Q + Q_2$ to hot.

→ Net: hot reservoir gains Q_2, cold loses Q_2, with no work input ⇒ Clausius violated.

4. Perpetual Motion Machine of Second Kind (PMM-II)

A heat engine that takes heat from a single reservoir and converts it completely into work, thereby having $\eta = 100\ \%$.

- It does not violate the first law (energy is conserved) but violates the second law (Kelvin-Planck).
- Example idea: extracting heat from ocean water to run a ship – impossible.
- So: PMM-I violates the first law; PMM-II violates the second law; both impossible.

5. Carnot's Principle (Carnot Theorems)

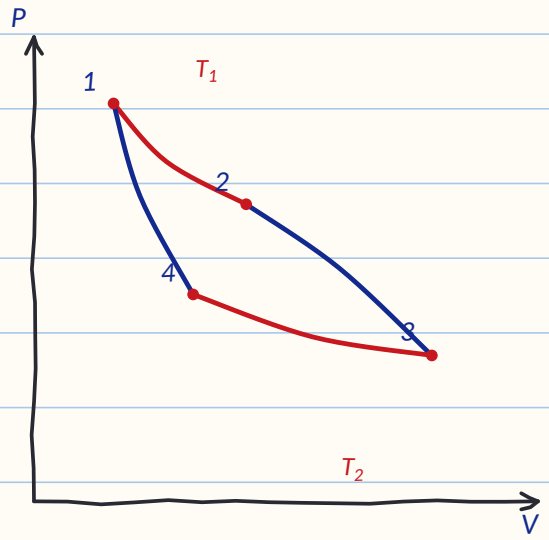
★ Note: Theorem 1: No heat engine operating between two given reservoirs can be more efficient than a reversible engine operating between the same two reservoirs.

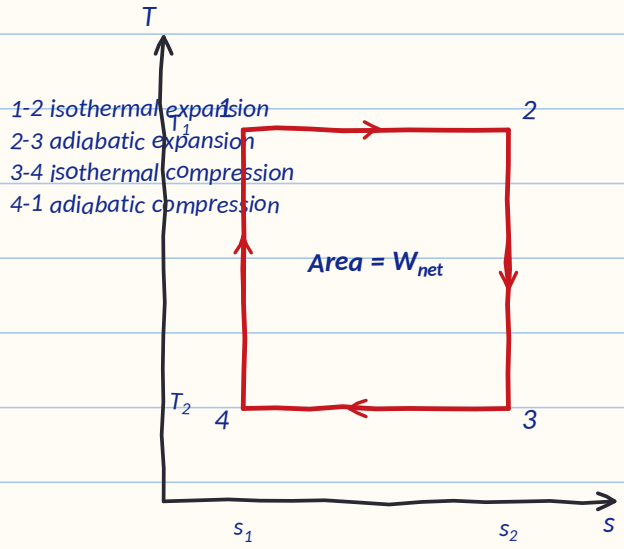
★ Note: Theorem 2: All reversible engines operating between the same two reservoirs have the same efficiency, whatever the working substance.

- Proof idea (Th.1): assume an irreversible engine has a higher efficiency; run a reversible engine as a heat pump with its work; combined device violates the Clausius / K-P statement.
- Hence $\eta_{rev} = f(T_1, T_2)$ only – leads to the thermodynamic temperature scale.

6. Carnot Cycle

A reversible cycle with two isothermal and two adiabatic (isentropic) processes, working between source T_1 and sink T_2.





Process	Nature	Heat / work
1-2	Isothermal expansion at T1	Q1 = mRT1 ln(V2/V1) absorbed
2-3	Adiabatic (isentropic) expansion	T falls T1→T2, work out
3-4	Isothermal compression at T2	Q2 = mRT2 ln(V3/V4) rejected
4-1	Adiabatic compression	T rises T2→T1, work in

From the adiabatic processes: $T_1/T_2 = (V_3/V_2)^{\gamma-1} = (V_4/V_1)^{\gamma-1} \Rightarrow V_2/V_1 = V_3/V_4$.

Hence the logarithmic terms cancel:

$$\eta_{Carnot} = W_{net}/Q_1 = 1 - Q_2/Q_1 = 1 - T_2/T_1$$

$$COP_{R,Carnot} = T_2/(T_1 - T_2) \qquad COP_{HP,Carnot} = T_1/(T_1 - T_2)$$

Specialties of the Carnot cycle

- It is completely reversible – the standard of perfection for all cycles.
- Its efficiency depends only on the reservoir temperatures, not on the working fluid.
- η increases when T_1 is raised or T_2 is lowered. η = 100 % only if T_2 = 0 K (impossible) or $T_1 = \infty$.
- The efficiency is the maximum possible between T_1 and T_2.
- Practical limits: needs infinitely slow isothermal heat transfer; very small work ratio; large V-ratio needed – hence not built in practice.

7. Thermodynamic (Kelvin) Scale of Temperature

Since η_{rev} depends only on temperatures, define the temperature ratio by the heat ratio of a reversible engine:

$$Q_1/Q_2 = T_1/T_2 \quad \text{(reversible engine)}$$

- Independent of any working substance or thermometric property – an *absolute* scale.
- Fixed point: triple point of water = 273.16 K. Then $T = 273.16\ (Q/Q_{tp})$.
- When $Q_2 \to 0$, $T_2 \to 0$: the absolute zero. Not attainable (third law).
- The ideal-gas scale coincides with the Kelvin scale.

8. Clausius Inequality

★ Note: For any cycle: $\oint \delta Q/T \le 0$. Equality holds for reversible cycles, strict inequality for irreversible cycles.

- Reversible cycle: $\oint \delta Q/T = 0$. Irreversible: $\oint \delta Q/T < 0$. Impossible cycle: $\oint \delta Q/T > 0$.
- Proof sketch: the cyclic system exchanges δQ at T with a Carnot engine linked to a reservoir T_0; apply K-P to the composite – net work cannot be positive $\Rightarrow \oint \delta Q/T \le 0$.
- Check for a cycle: $Q_1 = 1000$ kJ at 500 K, Q_2 rejected at 300 K. $\oint \delta Q/T = 2 - Q_2/300$. $Q_2 = 600 \Rightarrow 0$ (reversible); $700 \Rightarrow -0.33$ (irreversible, possible); $500 \Rightarrow +0.33$ (impossible).

9. Entropy

From $\oint (\delta Q/T)_{rev} = 0$, the quantity $\delta Q_{rev}/T$ is an exact differential – a property called entropy S.

$$dS = (\delta Q/T)_{rev} \qquad \Delta S = S_2 - S_1 = \int (\delta Q/T)_{rev}$$

- Unit kJ/K (extensive); specific entropy s in kJ/kg·K.
- Reversible adiabatic process: $dS = 0$ – isentropic. For irreversible processes: $dS > \delta Q/T$.

$$dS \geq \delta Q/T \quad \Rightarrow \quad \Delta S = \int \delta Q/T + S_{gen}, \quad S_{gen} \geq 0$$

Entropy change relations

$$\text{Ideal gas: } s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1)$$

$$s_2 - s_1 = c_p \ln(T_2/T_1) - R \ln(P_2/P_1)$$

- Isothermal heat transfer: $\Delta S = Q/T$. Phase change at T_{sat}: $\Delta s = h_{fg}/T_{sat}$. Liquid / solid: $\Delta s = c \ln(T_2/T_1)$.
- T-s diagram: area under a reversible curve = heat transfer $\int T\, ds$.

T ds relations

$$T\,ds = du + P\,dv \qquad T\,ds = dh - v\,dP$$

10. Principle of Entropy Increase

★ Note: The entropy of an isolated system (or of the universe) never decreases: $(\Delta S)_{isolated} \geq 0$; $\Delta S_{universe} = \Delta S_{system} + \Delta S_{surr} \geq 0$.

- > 0 irreversible process; $= 0$ reversible process; < 0 impossible.
- Entropy is not conserved; it is generated by irreversibilities (friction, finite ΔT heat transfer, mixing, free expansion).
- This is the quantitative form of the second law; gives the direction of natural processes.

Example: heat transfer through finite ΔT

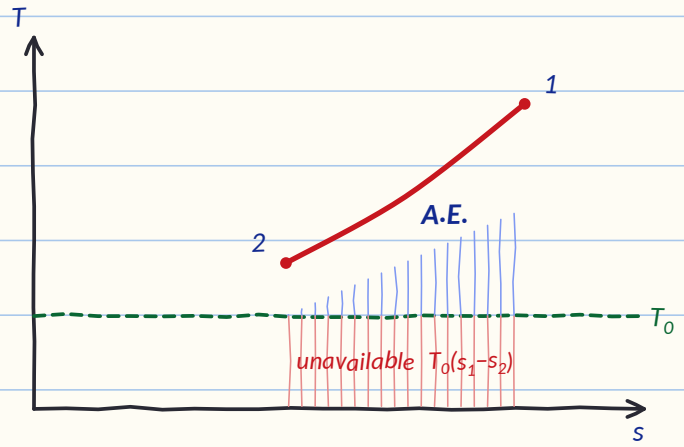
1000 kJ from 600 K body to 300 K body:

$\Delta S_{hot} = -1000/600 = -1.667$ kJ/K ; $\Delta S_{cold} = +1000/300 = +3.333$ kJ/K

$\Delta S_{univ} = +1.667$ kJ/K > 0 ⇒ irreversible

11. Availability and Irreversibility

Available energy (exergy): maximum useful work obtainable from a system as it comes to equilibrium with the environment (dead state: P_0, T_0). The rest is unavailable energy.



Available energy = $Q - T_0\Delta S$

Unavailable = $T_0\Delta S$

Max useful work from heat
→ between T and dead-state T_0

Heat from a constant-T source: $A.E. = Q\,(1 - T_0/T)$

Closed system: $\Phi = (U - U_0) + P_0(V - V_0) - T_0(S - S_0)$

Steady flow: $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + g\,z$

Irreversibility

$I = W_{rev} - W_{actual} = T_0\,S_{gen}$ $\qquad$ $\eta_{II} = W_{actual}/W_{rev}$

- Irreversibility = lost work (destroyed availability). Zero for a reversible process.
- Example: 1000 kJ at 600 K, T_0 = 300 K gives A.E. = 500 kJ. Transfer directly to a body at 400 K: $S_{gen} = 1000(1/400 - 1/600) = 0.833$ kJ/K ⇒ $I = 300 \times 0.833 = 250$ kJ lost.

- Second law efficiency compares actual performance to the best possible – it pinpoints where energy quality is wasted.

12. Thermodynamic Potentials – Helmholtz & Gibbs Functions

Helmholtz function $F = U - TS$

Gibbs function $G = H - TS$

Potential	Differential form	Natural variables
Internal energy U	$dU = T\,dS - P\,dV$	S, V
Enthalpy H	$dH = T\,dS + V\,dP$	S, P
Helmholtz F	$dF = -S\,dT - P\,dV$	T, V
Gibbs G	$dG = -S\,dT + V\,dP$	T, P

- Helmholtz: the maximum work obtainable from a closed system in a reversible isothermal process is $-\Delta F$ (at constant T, V: $dF \le 0$ for spontaneity).
- Gibbs: criterion of equilibrium at constant T and P: $dG \le 0$ spontaneous, $dG = 0$ equilibrium. During a phase change $g_f = g_g$.
- These are state functions (exact differentials), so cross-differentiation gives the Maxwell relations.

13. Maxwell Relations

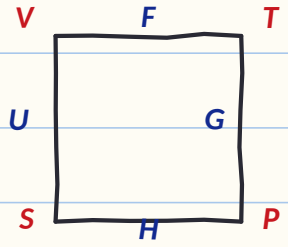
If $z = f(x, y)$ with $dz = M\,dx + N\,dy$, then $(\partial M/\partial y)_x = (\partial N/\partial x)_y$. Applying to the four potentials:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V \quad \text{[from } dU\text{]}$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P \quad \text{[from } dH\text{]}$$

$$(\partial S/\partial V)_T = (\partial P/\partial T)_V \qquad [\text{from } dF]$$

$$(\partial S/\partial P)_T = -(\partial V/\partial T)_P \qquad [\text{from } dG]$$



Edge letters = potentials: U, F, G, H

Corner letters = natural variables

Good Physicists Have Studied Under Very Fine Teachers (clockwise from top-left: V F T G P H S U)

- Use: replace hard-to-measure entropy derivatives by measurable P-V-T derivatives. Basis of the Clapeyron equation and of the Tds equations.

14. Third Law of Thermodynamics (Elementary)

★ Note: Nernst heat theorem / Third law: the entropy of a pure, perfectly crystalline substance is zero at absolute zero temperature.

- As $T \to 0$: $S \to 0$ and specific heats $\to 0$; this gives a reference for absolute entropy values.
- Absolute zero cannot be reached in a finite number of steps (unattainability principle).
- Heat capacity $c_p \to 0$ as $T \to 0$; thermal expansion coefficient $\to 0$.

15. Solved Numerical Problems

Problem 1 : Carnot efficiency

Engine between 800 K and 300 K: $\eta_{max} = 1 - 300/800 = 0.625 = 62.5\ \%$

Problem 2 : Carnot refrigerator

Cold space −10 °C (263 K), surroundings 27 °C (300 K):

$COP = 263/(300 - 263) = 7.11$

Problem 3 : entropy change of water

2 kg water heated 20 °C → 80 °C, $c = 4.18$ kJ/kg·K:

$\Delta S = m\, c\, \ln(T_2/T_1) = 8.36 \times \ln(353.15/293.15) = 1.56$ kJ/K

Problem 4 : isothermal expansion of air

1 kg air, volume doubles at constant T:

$\Delta S = m\, R\, \ln(V_2/V_1) = 0.287 \times \ln 2 = 0.199$ kJ/K

Problem 5 : Is the engine possible?

An engine receives 1000 kJ at 500 K and rejects 600 kJ at 300 K; claims to produce 400 kJ.

$\eta = 400/1000 = 40\ \%$ = Carnot limit $(1 - 300/500) \Rightarrow$ reversible, only just possible.

If it produced 500 kJ ($\eta = 50\ \%$) – impossible (> Carnot).

16. Entropy Changes in Ideal-Gas Processes

Process	Δs (per kg)
Isochoric (v const)	$c_v \ln(T2/T1) = c_v \ln(P2/P1)$
Isobaric (P const)	$c_p \ln(T2/T1) = c_p \ln(v2/v1)$
Isothermal	$R \ln(v2/v1) = -R \ln(P2/P1)$
Isentropic	0
Polytropic PV^n=C	$c_v(n-\gamma)/(n-1) \ln(T2/T1)$

- On the T-s diagram: isochoric lines are steeper than isobaric lines ($c_v < c_p$); isothermal = horizontal; isentropic = vertical.

17. Statistical Meaning of Entropy

- Boltzmann: $S = k \ln W$, where W = number of microstates for a given macrostate and $k = 1.38 \times 10^{-23}$ J/K.
- Entropy measures molecular disorder: ice → water → steam increases entropy; crystal at 0 K has $W = 1$, so $S = 0$ (third law).
- Natural processes move toward the most probable (most disordered) state; hence entropy of the universe increases – the "arrow of time".

18. First Law vs Second Law

First law	Second law
Conservation of energy	Quality and direction of energy
$Q - W = \Delta E$	$dS \geq \delta Q/T$
Defines internal energy U	Defines entropy S
Does not forbid PMM-II	Forbids PMM-II
Heat ⇌ work equivalence	Heat cannot be fully converted to work

Improving real-engine efficiency

→ Raise source temperature T_1 (limited by materials).

→ Lower sink temperature T_2 (limited by atmosphere / river water).

→ Reduce irreversibilities: friction, finite ΔT heat transfer, throttling, unrestrained expansion.

19. More Solved Problems

Problem 6 : free expansion

1 kg air, volume doubles by free expansion ($Q = 0$, $W = 0$):

$\Delta S_{sys} = 0.287 \ln 2 = 0.199$ kJ/K ; $\Delta S_{surr} = 0 \Rightarrow \Delta S_{univ} = +0.199$ kJ/K

Problem 7 : Carnot heat pump

Room 20 °C (293 K), outside 0 °C (273 K):

$COP_{HP} = 293/(293 - 273) = 14.65$

Problem 8 : Carnot engine

Output 30 kW between 600 K and 300 K: $\eta = 50\ \%$

$Q_1 = 30/0.5 = 60$ kW ; $Q_2 = 30$ kW

Problem 9 : melting ice

5 kg ice at 0 °C melts ($h_{if} = 334$ kJ/kg):

$\Delta S = m\, h_{if}/T = 5 \times 334/273.15 = 6.11$ kJ/K

Problem 10 : irreversibility of a real engine

Engine takes 100 kW at 900 K, produces 40 kW, rejects 60 kW at $T_0 = 300$ K.

$S_{gen} = 60/300 - 100/900 = 0.0889$ kW/K

$I = T_0 S_{gen} = 26.7$ kW ; $W_{rev} = 100(1 - 300/900) = 66.7$ kW

$\eta_{II} = 40/66.7 = 0.60$

20. Quick Revision & Important Questions

- K-P ⇒ no 100 % engine; Clausius ⇒ no free heat flow cold→hot; equivalent.
- $\eta_{Carnot} = 1 - T_2/T_1$; $\oint \delta Q/T \le 0$; $dS \ge \delta Q/T$; $\Delta S_{univ} \ge 0$.
- $F = U - TS$; $G = H - TS$; four Maxwell relations; $S(0\ K) = 0$.

→ State and prove the equivalence of Kelvin–Planck and Clausius statements.

→ Describe the Carnot cycle on P-V and T-s diagrams and derive its

efficiency.

→ State and prove the Clausius inequality. Define entropy.

→ What is availability? Derive irreversibility. Define second-law efficiency.

→ Derive Maxwell relations and explain Gibbs and Helmholtz functions.

~ End of Unit III ~